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Report on course work

«Optimal Irrigation Control in Cyber-Physical Agro Systems»

for the subject
«Challenges and Approaches of Modern Robotics»

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Contents

1 Motivation **3**

2 Research Problem Statement **4**
2.1 System Dynamics 4
2.2 Operational Constraints 4
2.3 Optimal Control Objective 5

3 Review of Related Approaches **5**
3.1 Reactive Heuristics and Classical Feedback Control 5
3.2 Model Predictive Control (MPC) 6
3.3 Artificial Intelligence of Things and Data-Driven Estimation 6

4 Baseline Methodology **7**
4.1 Justification of the Baseline 8
4.2 Baseline Control Architecture 8

5 Metrics for Evaluation **8**
5.1 Agronomic Performance and Constraint Satisfaction 9
5.2 Computational Latency at the Edge 9
5.3 Algorithmic Stability and Disturbance Rejection 9

6 Target Cyber-Physical Architecture **10**
6.1 The Edge Computing Layer (The Brain) 10
6.2 The Sensor Network (State Observation) 10
6.3 Robotic Actuation (Execution) 11

7 Conclusion **12**

1 Motivation

Water scarcity has emerged as one of the most critical environmental and socioeconomic challenges of the 21st century, particularly in arid and semi-arid regions such as Iran [1, 2]. The severity of this crisis is primarily driven by unsustainable patterns of human consumption rather than purely climatic factors [3, 4]. The agricultural sector accounts for approximately 90% of total freshwater withdrawals [4], operating within a geographical context where only a limited portion of land is naturally suited for intensive cultivation. To sustain production, agricultural practices have become heavily dependent on groundwater extraction, consistently exceeding natural recharge rates and leading to severe anthropogenic drought [3].

Efforts to address agricultural water use have traditionally focused on improving field-level irrigation efficiency through modern hardware technologies, such as sprinkler and drip systems. However, these approaches have not consistently resulted in reductions in overall water consumption. This contradiction is largely explained by the rebound effect, where gains in efficiency lead to behavioral and economic adjustments, such as the expansion of cultivated land or shifts to more water-intensive crops, that offset potential water savings [5]. Recent satellite-driven analyses of Iranian irrigation networks, such as the Doroodzan district, further expose massive systemic inefficiencies and conveyance losses that conventional management fails to mitigate [2]. Consequently, technological improvements alone are insufficient to guarantee meaningful water conservation.

This limitation highlights a fundamental issue: improving irrigation efficiency does not inherently enforce limits on total water consumption. As a result, there is a need for decision-making frameworks that explicitly account for system dynamics and resource constraints. This motivates the transition from purely hardware-based efficiency improvements toward data-driven and control-oriented irrigation strategies. The integration of sensing, computation, and actuation into a unified framework allows modern irrigation to operate as a Cyber-Physical System (CPS). Within this framework, agricultural irrigation, particularly in water-scarce regions, can be rigorously formulated as a constrained, dynamic control problem. Rather than treating irrigation as a static scheduling task, it is a sequential decision-making process operating under uncertainty, physical dynamics, and strict resource limitations. A defining feature of this problem is the presence of hard constraints, most notably a limited seasonal or annual water budget.

This thesis is motivated by the critical need to develop autonomous control policies capable of navigating these constraints. Solving the anthropogenic drought crisis requires closed-loop control algorithms that can optimize crop biomass yield while strictly adhering to finite water resource limits. By evaluating advanced control strategies, such as Reinforcement Learning (RL) and Model Predictive Control (MPC), against high-fidelity, spatially distributed environments, this research aims to evaluate the trade-offs between theoretically optimal control performance and computational feasibility in real-world, water-constrained agricultural systems.

2 Research Problem Statement

The fundamental problem addressed in this research is the formulation of a closed-loop autonomous irrigation policy capable of optimizing crop biomass yield while strictly adhering to finite water resource limits. For a uniform agricultural field, the soil-crop continuum is mathematically modeled as a lumped-parameter dynamical system.

2.1 System Dynamics

The state of the agricultural system is described by non-linear discrete-time difference equations at time step k. The primary state variable is the volumetric soil water content in the root zone, x_k, whose mass balance is governed by:

$$x_{k+1} = x_k + \frac{\Delta t}{Z_r}\Big(u_k + P_k - ET_k(x_k, w_k) - R_k(x_k) - D_k(x_k)\Big) \tag{1}$$

Where:

- Δt is the control sampling interval and Z_r is the effective root zone depth.
- u_k is the controllable control input (irrigation volume applied).
- P_k is the uncontrollable measured disturbance (precipitation).
- $ET_k(x_k, w_k)$ is the actual evapotranspiration, dependent on the current soil moisture state x_k and meteorological weather variables w_k.
- $R_k(x_k)$ is the surface runoff, triggered when water exceeds the soil infiltration rate.
- $D_k(x_k)$ is the deep drainage (percolation) below the root zone.

The secondary state variable is the cumulative crop biomass, B_k, which evolves dynamically based on the basal growth rate and is actively penalized by water stress conditions (drought or waterlogging) derived from x_k:

$$B_{k+1} = B_k + f_{growth}(x_k, w_k) \tag{2}$$

2.2 Operational Constraints

The optimal control policy $\pi(x_k) \to u_k$ must map the observed system states to an irrigation action while strictly satisfying physical actuators and agronomic safety boundaries:

1. **Agronomic State Bounds:** To prevent permanent wilting and severe waterlogging, the soil moisture state must remain bounded between the Wilting Point (WP) and Field Capacity (FC):

$$x_{WP} \leq x_k \leq x_{FC}, \quad \forall k \tag{3}$$

2. **Actuator Limits:** The irrigation volume applied at any time step is bounded by the maximum hydraulic capacity of the physical pumping system (U_{max}).

$$0 \leq u_k \leq U_{max}, \quad \forall k \tag{4}$$

3. **Resource Quota (Global Constraint):** The total water utilized over the entire growth season (prediction horizon T) is strictly limited by a finite seasonal water budget U_{quota}, reflecting regional water scarcity constraints:

$$\sum_{k=0}^{T} u_k \leq U_{quota} \tag{5}$$

2.3 Optimal Control Objective

The autonomous irrigation problem is formulated as a constrained optimal control problem. The objective is to find a policy π^* that minimizes the finite-horizon cost function J. The function is designed to penalize deviations from the maximum potential biomass (B_{max}) while simultaneously minimizing the total volume of irrigation water utilized:

$$\min_{\pi} J = \sum_{k=0}^{T-1} \left(Q\|B_{max,k} - B_k\|^2 + R\|u_k\|^2 \right) \tag{6}$$

Where Q and R are positive-definite weighting matrices that explicitly balance the agronomic profitability of the crop yield against the ecological and economic cost of water consumption.

3 Review of Related Approaches

The evolution of autonomous irrigation systems demonstrates a clear trajectory from reactive heuristic logic to predictive, multi-variable optimization. As agricultural requirements shift from simple yield maximization to strict resource conservation under climate uncertainty, the mathematical complexity of the underlying control architectures has scaled accordingly.

3.1 Reactive Heuristics and Classical Feedback Control

The earliest automated irrigation systems rely on open-loop scheduling or simple ON/OFF threshold logic (e.g., triggering irrigation when soil moisture drops below a static setpoint) [6]. While highly robust and computationally inexpensive, these systems are fundamentally reactive. They cannot anticipate unmeasured meteorological disturbances (such as imminent rainfall) or enforce long-term seasonal water quotas [7].

To achieve smoother continuous actuation, classical closed-loop controllers, such as Proportional-Integral-Derivative (PID) and Fuzzy Logic controllers, have been widely deployed [6]. However, the soil-crop continuum is a highly non-linear system characterized by massive time delays (the time between applying water and the water reaching the root zone) [7].

Classical controllers struggle to stabilize systems with large dead-times and lack the mathematical formulation required to handle strict, finite resource constraints over a prolonged seasonal horizon [7, 8].

3.2 Model Predictive Control (MPC)

To address the limitations of reactive control, the current state-of-the-art for constrained environments is Model Predictive Control (MPC). MPC utilizes a dynamic control-oriented model of the plant alongside meteorological forecasts to solve a constrained optimization problem over a receding prediction horizon.

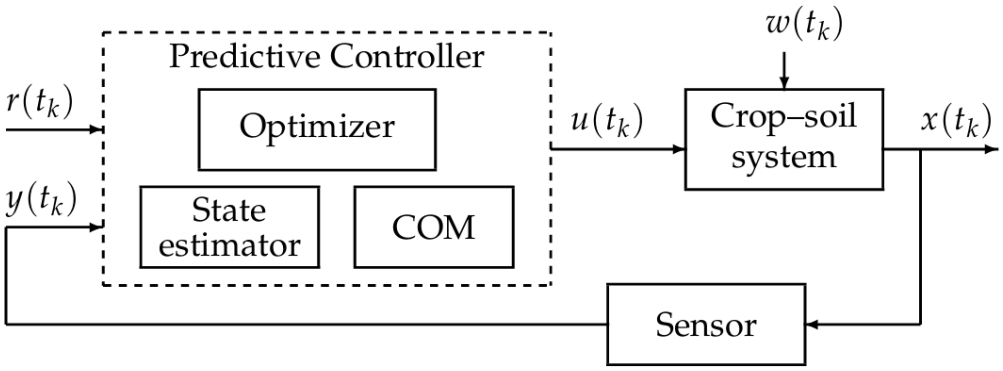

Figure 1: Architecture of a Model Predictive Controller, demonstrating the iterative online optimization process using a control-oriented plant model. Adapted from [8]

Recent literature highlights MPC as the benchmark for advanced irrigation. For example, frameworks have been developed that explicitly incorporate topographical runoff and crop-soil mass balances to govern irrigation policies under strict physical constraints [8]. The primary advantage of MPC is its ability to mathematically guarantee constraint satisfaction (e.g., never exceeding the seasonal water budget). However, its fundamental drawback is the immense computational burden required to solve non-linear optimization matrices online at every sampling step, severely limiting its deployment on low-power, decentralized edge computing hardware.

3.3 Artificial Intelligence of Things and Data-Driven Estimation

To overcome the computational bottlenecks of mechanistic models, recent trends have shifted towards Artificial Intelligence of Things (AIoT) and data-driven methods [9]. As highlighted in comprehensive reviews of next-generation crop modeling, the scientific community is actively calling for hybrid frameworks that combine mechanistic physical foundations with AI to enhance real-time decision support [10].

Currently, much of the AI application in smart irrigation focuses on supervised machine learning and predictive IoT architecture for open-loop estimation. For instance, recent frameworks utilize Recurrent Neural Networks (RNN-LSTM) to accurately predict subsurface soil moisture dynamics [11], while other implementations employ self-adaptive incremental machine learning on edge devices to trigger drip irrigation based on real-time environmental data [12]. Similarly, ensemble learning methods, such as Random Forest algorithms optimized via swarm

intelligence, are increasingly used to predict real-time irrigation requirements based on massive multi-source meteorological datasets [13].

While these models achieve exceptional predictive accuracy, they function primarily as open-loop estimators; they lack the mathematical architecture to natively enforce physical system constraints or calculate optimal control actions over a multi-stage horizon. These limitations motivate further investigation into whether learning-based control methods can operate reliably under strict irrigation constraints.

4 Baseline Methodology

This research aims to systematically evaluate whether learning-based control approaches, such as Reinforcement Learning (RL), can approximate the performance of constraint-aware model-based methods such as Model Predictive Control (MPC) in water-constrained irrigation systems.

RL provides a framework for sequential decision-making by learning control policies through interaction with a dynamic environment. Unlike supervised learning methods, which are primarily used for open-loop prediction, RL agents (e.g., Deep Deterministic Policy Gradient (DDPG) and Soft Actor-Critic (SAC)) optimize a reward function to generate control actions. By training these policies offline against high-fidelity mechanistic models, the computational burden of optimization can be largely shifted to the training phase, enabling fast online execution.

However, the application of RL to irrigation control remains limited and insufficiently explored, particularly in environments characterized by strict physical and resource constraints. Most existing applications of machine learning in agriculture focus on prediction tasks (e.g., soil moisture estimation or irrigation demand forecasting) rather than closed-loop control policy design.

A key limitation of standard RL methods is the lack of inherent mechanisms for enforcing hard constraints. In irrigation systems, violations of constraints, such as exceeding water quotas or inducing soil saturation, can have irreversible agronomic and environmental consequences. While recent developments in constrained and safe RL aim to address these challenges, their application to realistic agro-hydrological systems remains limited.

This leads to a fundamental trade-off: RL offers scalability and low-latency decision-making after training, but may lack the reliability and constraint guarantees provided by model-based approaches such as MPC. Consequently, it remains unclear whether RL can serve as a viable alternative for constrained irrigation control.

To enable a rigorous comparison, a high-performance model-based control framework is required as a reference benchmark. The integration of optimal control algorithms with high-fidelity Agent-Based Models (ABMs) remains a well-documented computational challenge across modern cyber-physical systems [14]. Because ABMs are computationally intractable for online optimization, a highly rigorous benchmark is required to test if RL can safely bypass this bottleneck. Therefore, this research adopts the **Agent-Based Model Predictive Control**

(ABM-MPC) framework proposed by Lopez-Jimenez et al. (2024) as the rigorous mathematical baseline.

4.1 Justification of the Baseline

For a large-scale agricultural environment (e.g., a 16-hectare plot), treating the field as a single lumped-parameter system yields significant modeling errors due to spatial heterogeneity. Water runoff pools in lower elevations, causing simultaneous localized drought and waterlogging across the same field. The selected baseline explicitly addresses this by employing an Agent-Based Model (ABM) as the simulation-oriented model (SOM), dynamically computing the topographical water fluxes (ϕ_4) between discrete 100m $\times$ 100m soil-crop agents.

4.2 Baseline Control Architecture

The baseline utilizes a central Model Predictive Controller. At each discrete time step k, the MPC receives updated state estimations for all N agents. It utilizes a reduced-order control-oriented model (COM) alongside meteorological weather forecasts to construct a multi-variable optimization matrix. The MPC solver computes the optimal continuous irrigation volumes for all N valves over a receding prediction horizon ($H_p = 14$ days), strictly bounded by the physical actuator limits and the global seasonal water quota.

While this ABM-MPC framework represents a high-performance benchmark for constraint-aware irrigation control, calculating the non-linear objective function for 80+ interconnected agents requires immense computational power. Therefore, this baseline serves a dual purpose in this thesis:

1. It acts as the "Virtual Plant" (simulation environment) used to train the RL agent offline.
2. It provides a reference benchmark for both biomass yield and computational performance under constraint-aware control.

To overcome the computational bottleneck of the baseline, this research evaluates three state-of-the-art continuous-control Reinforcement Learning algorithms as candidates: Deep Deterministic Policy Gradient (DDPG), Twin Delayed DDPG (TD3), and Soft Actor-Critic (SAC). Following a preliminary algorithmic assessment, the single most suitable candidate will be selected and fully implemented. The chosen RL policy will be tasked with mapping the exact same state observations to valve actuations, with the hypothesis that it will match the MPC's agronomic performance while shifting the computational burden largely offline.

5 Metrics for Evaluation

The comparative success of the implemented RL control policy against the MPC baseline will be evaluated across three primary engineering domains to ensure both theoretical robustness and practical deployment viability:

5.1 Agronomic Performance and Constraint Satisfaction

The fundamental objective of the autonomous controller is to maximize agricultural yield without violating strict resource limits, directly addressing the macro-level challenge of severe water scarcity. This metric evaluates the ability of the chosen RL policy (π_{RL}) to maintain the soil moisture state variable (x_k) within safe agronomic bounds while strictly enforcing the regional seasonal water quota (U_{quota}). Mathematically, the policy must satisfy:

$$x_{WP} \leq x_k \leq x_{FC} \quad \forall k \in \{0, \ldots, T\} \tag{7}$$

$$\sum_{k=0}^{T-1} u_k^{(RL)} \leq U_{quota} \tag{8}$$

Where x_{WP} is the Wilting Point and x_{FC} is the Field Capacity. Performance will be quantitatively measured by comparing the total normalized biomass accumulated at harvest (B_T) and the total volume of irrigation applied against the theoretical optimum established by the MPC baseline over a complete finite crop growth simulation horizon ($T = 120$ days).

5.2 Computational Latency at the Edge

To resolve the primary deployment bottleneck for modern cyber-physical agricultural systems, the trained policy must be lightweight enough to execute on resource-constrained IoT microcontrollers embedded within agricultural robotics.

By its fundamental mathematical nature, MPC calculates the non-linear objective function online at every sampling step, making its execution inherently computationally expensive. Conversely, RL shifts the heavy optimization to the offline learning phase, leaving only lightweight feed-forward neural network execution for the online policy. Therefore, the inequality $\bar{\tau}_{RL} \ll \bar{\tau}_{MPC}$ is a known theoretical consequence.

This metric explicitly quantifies this expected latency reduction. Let τ_k denote the execution time required to compute the control action u_k at time step k. The computational feasibility will be rigorously profiled by tracking the average execution time ($\bar{\tau}$) and the worst-case maximum latency (τ_{max}) across the simulation steps K:

$$\bar{\tau} = \frac{1}{K} \sum_{k=1}^{K} \tau_k, \quad \tau_{max} = \max_k(\tau_k) \tag{9}$$

Quantifying this margin validates the algorithm's specific suitability for real-time edge execution on mobile robotic hardware, where power and processing overhead are strictly limited.

5.3 Algorithmic Stability and Disturbance Rejection

To ensure the autonomous policy behaves safely and predictably when deployed in highly stochastic, real-world climates, it must demonstrate robust disturbance rejection. This metric evaluates the agent's ability to handle unpredicted meteorological noise without catastrophic failure.

Algorithmic stability will be quantified by subjecting the closed-loop system to an unmeasured stochastic disturbance vector d_k, such that the actual meteorological input is $w_k = \hat{w}_k + d_k$, where d_k represents extreme unforecasted rainfall variance ($\pm 20\%$). The robustness of the RL controller will be measured by evaluating the magnitude of the state tracking error (e_k) and calculating the frequency of physical constraint violations ($V_{penalty}$) over the disturbance horizon:

$$e_k = |x_k^{(RL)} - x_{ref}| \tag{10}$$

$$V_{penalty} = \sum_k \mathbf{1}(x_k < x_{WP} \vee x_k > x_{FC}) \tag{11}$$

Where $\mathbf{1}(\cdot)$ is the indicator function for boundary violations. Minimizing these mathematically guarantees that the AI policy maintains safe operational boundaries even when the environmental simulation severely deviates from the nominal training data.

6 Target Cyber-Physical Architecture

To bridge the gap between theoretical optimal control and real-world agricultural deployment, the proposed Reinforcement Learning (RL) policy must be integrated into a functional Cyber-Physical System (CPS). Modern smart agriculture heavily relies on the Internet of Things (IoT) to seamlessly connect physical environmental states with digital decision-making algorithms [15]. Within this architecture, the control framework is structured across three distinct physical layers: edge computation, state observation, and robotic actuation.

6.1 The Edge Computing Layer (The Brain)

A critical challenge in smart agriculture is the reliance on cloud-based computing, which frequently suffers from high latency, packet loss, and poor connectivity in remote, rural environments [16]. While traditional Model Predictive Control (MPC) often requires cloud-level processing to solve complex multi-variable optimization matrices, the proposed RL architecture explicitly mitigates this bottleneck.

Because the computationally intensive optimization of the RL agent occurs offline during the training phase, the resulting policy is a lightweight feed-forward neural network. This allows the "brain" of the control system to be deployed directly onto edge computing devices (e.g., Raspberry Pi or NVIDIA Jetson microcontrollers) embedded locally on the farm or onboard mobile agricultural robots. Edge deployment ensures high-speed, localized inference, directly satisfying the computational latency metric ($\bar{\tau}_{RL} \ll \bar{\tau}_{MPC}$) and enabling the system to operate continuously even during network outages [17].

6.2 The Sensor Network (State Observation)

To calculate the optimal irrigation action, the edge-deployed RL agent requires continuous, real-time feedback regarding the soil moisture state (x_k) and meteorological disturbances (w_k). This state observation is achieved through a distributed, multi-modal sensing network:

- **In-Situ IoT Sensors:** Wireless Sensor Networks (WSNs) utilizing low-power wide-area network (LPWAN) protocols (e.g., LoRaWAN) transmit continuous subsurface soil moisture and volumetric water content data to the edge controller [18].

- **Aerial State Estimation:** To account for spatial heterogeneity across large plots, Unmanned Aerial Vehicles (UAVs) equipped with thermal and multispectral cameras can serve as mobile sensor platforms. UAV imagery provides high-resolution data regarding crop water stress and topographical drainage pooling, mapping the discrete state variables across the Agent-Based Model grid [17].

Figure 2: The DJI Mavic 3 Multispectral Drone utilized for high-resolution crop water stress mapping.

6.3 Robotic Actuation (Execution)

Once the RL policy calculates the optimal constrained control action (u_k), the command is passed to the physical actuation layer for execution. By treating the calculated irrigation volume (u_k) as a localized payload command, the algorithm natively supports both stationary and mobile robotic deployment:

- **Stationary Smart Valves:** In traditional precision drip or sprinkler setups, the control command regulates automated solenoid valves, adjusting the flow rate dynamically based on the algorithmic quota [15].

- **Autonomous Mobile Robots (AGVs):** For advanced robotic deployments, the Agent-Based Model grid functions as a navigation coordinate map. Unmanned Ground Vehicles navigate autonomously between crop rows. Upon reaching the designated grid cells, the robot executes the exact u_k water payload directly to the crop root zone, minimizing surface evaporation and bypassing the need for extensive permanent pipe infrastructure [18, 16].

Figure 3: The 360 Rain autonomous irrigation robot executing localized u_k water payload commands.

7 Conclusion

This research bridges the critical gap between macroeconomic water conservation challenges and advanced cyber-physical control theory. By mathematically formulating the anthropogenic drought crisis as a constrained optimal control problem, this thesis proposes an active transition away from reactive, hardware-centric irrigation toward predictive, data-driven autonomy.

While Model Predictive Control (MPC) offers mathematical guarantees for satisfying strict resource quotas, its immense computational complexity fundamentally limits its deployment on the resource-constrained edge hardware typical of mobile agricultural robots and AIoT sensor networks. By exploring continuous-control Reinforcement Learning (RL) as an alternative, this research aims to determine if the computational burden of constrained multi-variable control can be safely shifted to an offline training phase.

Ultimately, this work seeks to validate a closed-loop AI control architecture that achieves the agronomic safety and yield optimization of state-of-the-art MPC, while maintaining the millisecond-latency execution required for real-time deployment on scalable, real-world robotic irrigation networks.

References

[1] R. Behling, S. Roessner, and S. Foerster, "Interrelations of vegetation growth and water scarcity in iran revealed by satellite time series," *Scientific Reports*, vol. 12, p. 20784, 2022.

[2] P. Afrasiabikia, A. Parvaresh Rizi, and L. Brocca, "A satellite-driven approach to estimating delivered irrigation water via sm-based inversion algorithm: A case study of the doroodzan irrigation district, iran," *Agricultural Water Management*, vol. 325, p. 110141, 2026.

[3] M. B. Mesgaran and P. Azadi, "A national adaptation plan for water scarcity in iran," Stanford Iran 2040 Project, Working Paper 6, 2018.

[4] H. Nouri *et al.*, "Water management dilemma in the agricultural sector of iran: A review focusing on water governance," *Agricultural Water Management*, vol. 278, p. 108162, 2023.

[5] C. D. Pérez-Blanco, C. H. Hampson, and C. G. Coggan, "Irrigation technology and water conservation: A review of the theory and evidence," *Review of Environmental Economics and Policy*, vol. 14, no. 2, pp. 216–239, 2020.

[6] V. Sharma, G. Kaur, S. S, V. Chhabra, and R. Kashyap, "Smart irrigation systems in agriculture: An overview," *Computers and Electronics in Agriculture*, vol. 239, p. 111008, 2025.

[7] J. Lopez-Jimenez, L. Dewasme, A. Vande Wouwer, and N. Quijano, "Dynamic modeling of crop-soil systems to design monitoring and automatic irrigation processes: A review with worked examples," *Water*, vol. 14, no. 15, p. 2404, 2022.

[8] J. Lopez-Jimenez, N. Quijano, L. Dewasme, and A. Vande Wouwer, "Agent-based model predictive control of soil-crop irrigation with topographical information," *Control Engineering Practice*, vol. 150, p. 106012, 2024.

[9] J. Bayar, N. Ali, Z. Cao, Y. Ren, and Y. Dong, "Artificial intelligence of things (aiot) for precision agriculture: applications in smart irrigation, nutrient and disease management," *Smart Agricultural Technology*, vol. 12, p. 101629, 2025.

[10] A. S. Mgana, A. Huo, Y. Zhao, P. R. Joshi, and A. Firaouni, "Advancing crop growth modeling: Challenges, model integration, and pathways for improved decision support," *Ecological Modelling*, vol. 512, p. 111415, 2026.

[11] S. Maniam, Y.-K. Tee, E. Memar, H. Y. Wong, and M. Zaman, "Smart irrigation management: Iot-based rnn-lstm model for soil moisture prediction in precision agriculture," *Smart Agricultural Technology*, vol. 13, p. 101866, 2026.

[12] S. Mahmud, M. M. Rahaman, S. A. Rafiq, F. A. Antara, T. Tasnim, M. M. Rahman, and S. Enam, "Self-adaptive incremental learning-based real-time smart drip irrigation management using iot and mobile application," *Smart Agricultural Technology*, vol. 13, p. 101858, 2026.

[13] W. Hu, Y. Liu, J. An, S. Xu, Z. Zhou, M. An, X. Guo, X. Ma, W. Jiang, and Y. Wang, "Intelligent irrigation strategy model for farmland using dung beetle optimization-random forest algorithms," *Agricultural Water Management*, vol. 317, p. 109653, 2025.

[14] P. Polcz, I. Z. Reguly, K. Tornai, J. Juhász, S. Pongor, and A. Csikász-Nagy, "Smart epidemic control: A hybrid model blending odes and agent-based simulations for optimal, real-world intervention planning," *PLoS Computational Biology*, vol. 21, no. 5, p. e1013028, 2025.

[15] G. Farrokhi and M. Gapeleh, "Smart agriculture based on internet of things," *Roshd-e-Fanavari (Technology Incubator Quarterly)*, vol. 15, no. 59, pp. 30–37, 2019.

[16] F. Hesabi, I. Jafarpanah, A. Karshenas, and M. Ahmadzadeh, "Technological trends in smart agriculture," *Journal of Entrepreneurial Strategies in Agriculture*, vol. 10, no. 2, pp. 100–115, 2023.

[17] M. Pourgholam-Amiji, I. Hajirad, K. Ahmadaali, and A. Liaghat, "Smart irrigation based on the iot," *Iranian Journal of Soil and Water Research*, vol. 55, no. 9, pp. 1647–1678, 2024.

[18] I. Hajirad, "Identification of smart irrigation management methods and tools and their impact on optimizing irrigation water consumption," *Journal of Water and Sustainable Development*, vol. 12, no. 2, pp. 159–176, 2025.